Lecture 4
Multi-state quantum systems: Composition and evolution

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What we have learned so far?

- We have introduced the notion of a qubit, the quantum analog to a bit, as the prototypical example of a single quantum state.
- A qubit is typically represented as a superposition, i.e., a linear combination of normalised and mutually orthogonal quantum states labelled by $|0\rangle$ and $|1\rangle$ with complex coeficients:

$$|v\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

- and we also learned that mathematically such states live in a Hilbert space, i.e., a complex vector space with internal products.
- Finally, we have seen that qubits evolve through the application of unitary operations, e.g., the Hadamard gate, and can also be measured.

The new questions

Questions

- How can bigger quantum states be formed from smaller ones?
- How do such systems evolve and be measured?

New states from older

States (classical, probabilistic, quantum) are represented by vectors in a specific vector space. Thus, putting two such states side by side

$$|v\rangle|u\rangle$$

corresponds to computing their tensor product: $|v\rangle \otimes |u\rangle$.

Example

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \otimes \begin{bmatrix} d \\ e \\ f \end{bmatrix} = \begin{bmatrix} a \begin{bmatrix} d \\ e \\ f \end{bmatrix} \\ b \begin{bmatrix} d \\ e \\ f \end{bmatrix} \\ c \begin{bmatrix} d \\ e \\ f \end{bmatrix} \end{bmatrix} = \begin{bmatrix} ad \\ ae \\ af \\ bd \\ be \\ bf \\ cd \\ ce \\ cf \end{bmatrix}$$

Note

Some convention is needed to encode the joint classical states as row indices, typically the lexicographic order.

Tensor product $V \otimes W$

$\otimes$ is bilinear

- $(|u_1\rangle + |u_2\rangle)\otimes|z\rangle \;=\; |u_1\rangle\otimes|z\rangle + |u_2\rangle\otimes|z\rangle$
- $\alpha|u\rangle\otimes|z\rangle \;=\; |u\rangle\otimes\alpha|z\rangle \;=\; \alpha(|u\rangle \otimes |z\rangle)$
- and similarly in the second argument

Thus, scalars float freely over $\otimes$:

$$\alpha|u_1\rangle\otimes|u_2\rangle \;=\; |u_1\rangle\otimes\alpha|u_2\rangle \;=\; \alpha(|u_1\rangle\otimes|u_2\rangle)$$

Notation

$$|uv\rangle = |u\rangle|v\rangle = |u\rangle \otimes |v\rangle$$

New states from older

The Composition Postulate

The state space of a combined (quantum) system is the tensor product $V \otimes W$ of the state spaces V and W of its components.

Note

n m-dimensional vectors $\rightsquigarrow$ a vector in m^n-dimensional space

i.e., the state space of a quantum system grows exponentially with the number of particles.

Composing probabilistic states

The composition of two probabilistic systems X and Y over a set of classical states A and B yields

$$|w\rangle = \sum_{(a,b)\in A\times B} p_{(a,b)}(|a\rangle \otimes |b\rangle) = \sum_{(a,b)\in A\times B} p_{(a,b)}|ab\rangle$$

Independence

If observing one system does not have any effect on the probabilities associated with the other, the systems are independent, i.e., the probability of observing a particular joint state is the product of probabilities of observing the corresponding value in each state, i.e.,

$$\Pr((X,Y)=(a,b)) = \Pr(X=a)\Pr(Y=b)$$

Or, in Dirac's notation, there exists states $|v\rangle = \sum_{a\in A} p_a|a\rangle$ and $|u\rangle = \sum_{b\in B} p_b|b\rangle$ st, for all $a \in A, b \in B$,

$$p_{(a,b)} = p_a p_b$$

Composing probabilistic states

Independent states

Are the ones obtained compositionally as a tensor product, i.e.,

$$|w\rangle \;=\; |u\rangle \otimes |v\rangle \;=\; \sum_a \sum_b p_a p_b |ab\rangle$$

for $|v\rangle = \sum_{a\in A} p_a|a\rangle$ and $|u\rangle = \sum_{b\in B} p_b|b\rangle$.

Or, in an equivalent formulation,

$$\langle ab|w\rangle \;=\; \langle a|v\rangle\langle b|u\rangle$$

for all $a \in A, b \in B$.

Measuring a composed state

If only system X is measured with outcome a,

- $\Pr(X = a) = \sum_{b \in B} p_{ab}$ (marginal probability of observing a)
- Thus, the probabilistic state associated with X is

$$\sum_a (\sum_{b \in B} p_{ab}) |a\rangle$$

- The outcome of a future measurement of Y remains uncertain, which is captured by the notion of conditional probability:

$$\Pr(Y = b) | (X = a) = \frac{\Pr((X, Y) = (a, b))}{\Pr(X = a)}$$

- accordingly, the corresponding probabilistic state becomes

$$|a\rangle |w_a\rangle \quad \text{where} \quad |w_a\rangle = \frac{\sum_{b \in B} p_{ab} |b\rangle}{\sum_{b \in B} p_{ab}}$$

Operating over a composed state

Acting independently on individual states

The composed operation is the tensor product of the (matrices of the) individual ones:

$$M \otimes N = \begin{bmatrix} m_{11}N & m_{12}N & \cdots & m_{1n}N \\ m_{21}N & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ m_{n1}N & \cdots & \cdots & \cdots \end{bmatrix}$$

Practice

Compute the matrix that applies X and Id to the first and the second qubit, respectively.

Operating over a composed state

But often this is not the case ...

Practice

Compute the matrix corresponding to performing one of the two operations below with a probability of 50%:

- Set the second qubit equal to the first.
- Set the first qubit equal to the second.

$$\frac{1}{2}\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Composing quantum states

The setting is similar to the probabilistic case.

Quantum states of multiple systems are represented by column vectors with complex entries and norm equal to 1, just as in the single-state case.

Example

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{6}} \\ \frac{i}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}$$ which corresponds to

$$\frac{1}{\sqrt{2}}|00\rangle - \frac{1}{\sqrt{6}}|01\rangle + \frac{i}{\sqrt{6}}|10\rangle + \frac{1}{\sqrt{6}}|11\rangle$$

Composing quantum states

Composing quantum states with $\otimes$ generates quantum states. The basic observation is that norm is multiplicative with respect to tensor:

$$\begin{aligned} \||u\rangle \otimes |v\rangle| &= \sqrt{\sum_{(a,b)\in A\times B} |\langle ab|u\otimes v\rangle|^2} \\ &= \sqrt{\sum_{a\in A}\sum_{b\in B} |\langle a|u\rangle\langle b|v\rangle|^2} \\ &= \sqrt{\sum_{a\in A} |\langle a|u\rangle|^2 \sum_{b\in B} |\langle b|v\rangle|^2} \;=\; |u||v| \end{aligned}$$

Clearly, $\||u\rangle \otimes |v\rangle| = 1$, given that $\||u\rangle| = \||v\rangle| = 1$.

Composing quantum states

Example

The tensor product of $\alpha_1|0\rangle + \alpha_2|1\rangle$ and $\beta_1|0\rangle + \beta_2|1\rangle$ is

$$\alpha_1\beta_1|0\rangle \otimes |0\rangle \ + \ \alpha_1\beta_2|0\rangle \otimes |1\rangle \ + \ \alpha_2\beta_1|1\rangle \otimes |0\rangle \ + \ \alpha_2\beta_2|1\rangle \otimes |1\rangle$$

or, in a simplified notation,

$$\alpha_1\beta_1|00\rangle \ + \ \alpha_1\beta_2|01\rangle \ + \ \alpha_2\beta_1|10\rangle \ + \ \alpha_2\beta_2|11\rangle$$

Note

Notice the set $\{|0\rangle \otimes |0\rangle, |0\rangle \otimes |1\rangle, |1\rangle \otimes |0\rangle, |1\rangle \otimes |1\rangle\}$ is a basis for the space of two qubits.

Independent and entangled states

As in the probabilistic case, tensoring quantum states yields independent multi-state systems.

However, most multi-state systems are entangled, which is itself a basic resource for quantum computation:

Examples

$$\begin{aligned} \mathrm{CHZ} &= \frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|111\rangle \\ \mathrm{W} &= \frac{1}{\sqrt{3}}|001\rangle + \frac{1}{\sqrt{3}}|010\rangle + \frac{1}{\sqrt{3}}|100\rangle \end{aligned}$$

A note on representing multi-qubit states

Any unit vector in a 2^n Hilbert space represents a possible n-qubit state, up to

... a certain level of redundancy

- As before, vectors that differ only in a global phase represent the same quantum state
- but also the same phase factor in different qubits of a tensor product represent the same state:

$$|u\rangle \otimes (e^{i\Phi}|v\rangle) \;=\; e^{i\Phi}(|u\rangle \otimes |v\rangle) \;=\; (e^{i\Phi}|u\rangle) \otimes |v\rangle$$

Actually, phase factors in qubits of a single term of a superposition can always be factored out into a coefficient for that term, i.e. phase factors distribute over tensors

Partial measurements

Similarly to the probabilistic case, the quantum state corresponding to the aggregation of two systems takes the form of a unit vector

$$|u\rangle = \sum_{(a,b)\in A\times B} \alpha_{ab}|ab\rangle$$

When measured, the probability of obtaining (a, b) is given by

$$|\langle ab|u\rangle|^2 = |\alpha_{ab}|^2$$

If only the first system is measured, the probability of getting a is

$$\sum_{b\in B} |\langle ab|u\rangle|^2 = \sum_{b\in B} |\alpha_{ab}|^2$$

Question

What happens to the state of the second system?

Partial measurements

First observe that

$$|u\rangle \;=\; \sum_{a\in A} |a\rangle \otimes |u_a\rangle \quad \text{where} \quad |u_a\rangle \;=\; \sum_{b\in B} \alpha_{ab}|b\rangle$$

Then, the probability of obtaining a is given by

$$\sum_{b\in B} |\alpha_{ab}|^2 \;=\; \||u_a\rangle|^2$$

and the quantum state becomes

$$|a\rangle \otimes \frac{|u_a\rangle}{\||u_a\rangle|}$$

i.e., the state collapses consistently with the outcome a.

Note

There is, however, no analogous to the marginal probabilistic state: $\sum_{(a,b)\in A\times B} \alpha_{ab}|a\rangle$ is not, in general, a quantum state.

Partial measurements

Example

Let

$$|u\rangle = \frac{1}{\sqrt{2}}|00\rangle - \frac{1}{\sqrt{6}}|01\rangle + \frac{i}{\sqrt{6}}|10\rangle + \frac{1}{\sqrt{6}}|11\rangle$$

Start rewriting $|u\rangle$ as

$$|u\rangle = |0\rangle \otimes \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{6}}|1\rangle\right) + |1\rangle \otimes \left(\frac{i}{\sqrt{6}}|0\rangle + \frac{1}{\sqrt{6}}|1\rangle\right)$$

The probability associated with outcome 0 is

$$\left|\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{6}}\right|^2 = \frac{2}{3}$$

and the resulting state

$$|0\rangle \otimes \frac{\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{6}}|1\rangle}{\sqrt{\frac{2}{3}}} = |0\rangle \otimes \left(\frac{\sqrt{3}}{2}|0\rangle - \frac{1}{2}|1\rangle\right)$$

Bases

The computational basis for a vector space

$$\underbrace{V \otimes V \otimes \cdots \otimes V}_{n}$$

corresponding to the composition of n qubits (each living in V) is the set

$$\{\underbrace{|0\rangle \cdots |0\rangle|0\rangle}_{n}, \underbrace{|0\rangle \cdots |0\rangle|1\rangle}_{n}, \underbrace{|0\rangle \cdots |1\rangle|0\rangle}_{n}, \cdots \underbrace{|1\rangle \cdots |1\rangle|1\rangle}_{n}\}$$

$$\overset{\text{abv}}{=}$$

$$\{\underbrace{|0\cdots 00\rangle}_{n}, \underbrace{|0\cdots 01\rangle}_{n}, \underbrace{|0\cdots 10\rangle}_{n}, \cdots \underbrace{|1\cdots 11\rangle}_{n}\}$$

which may be written in a compressed (decimal) way as

$$\{|0\rangle, |1\rangle, |2\rangle, |3\rangle, \cdots |2^n - 1\rangle\}$$

Bases

Example

The computational basis for a two qubit system is

$$\{|0\rangle, |1\rangle, |2\rangle, |3\rangle\}$$

with

$$|0\rangle = |00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad |1\rangle = |01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad |2\rangle = |10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad |3\rangle = |11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Bases

There are of course other bases ... besides the standard one, e.g.

The Bell basis

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$
$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$
$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$
$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Compare with the Hadamard basis for the single qubit systems

Computing with quantum states

If a quantum state is a ray (i.e. a unit vector in a Hilbert space H up to a global phase), its evolution is specified a certain kind of linear operators

$$U : H \longrightarrow H$$

Linearity

$$U\left(\sum_j \alpha_j |v_j\rangle\right) = \sum_j \alpha_j U(|v_j\rangle)$$

just by itself has an important consequence:

unknown quantum states cannot be cloned

The no-cloning theorem

Linearity implies that unknown quantum states cannot be cloned

Let $U(|a\rangle|0\rangle) = |a\rangle|a\rangle$ be a 2-qubit operator and $|c\rangle = \frac{1}{\sqrt{2}}(|a\rangle + |b\rangle)$ for $|a\rangle$, $|b\rangle$ orthogonal. Then,

$$\begin{aligned} U(|c\rangle|0\rangle) &= \frac{1}{\sqrt{2}}(U(|a\rangle|0\rangle) + U(|b\rangle|0\rangle)) \\ &= \frac{1}{\sqrt{2}}(|a\rangle|a\rangle + |b\rangle|b\rangle) \\ &\neq \frac{1}{\sqrt{2}}(|a\rangle|a\rangle + |a\rangle|b\rangle + |b\rangle|a\rangle + |b\rangle|b\rangle) \\ &= |c\rangle|c\rangle \\ &= U(|c\rangle|0\rangle) \end{aligned}$$

As already seen, $|x\rangle|y\rangle = |xy\rangle = |x\rangle \otimes |y\rangle$

The state evolution postulate

State Evolution Postulate

The evolution over time of the state of a closed quantum system is described by a unitary operator.

The evolution is linear

$$U\left(\sum_j \alpha_j \left|v_j\right\rangle\right) = \sum_j \alpha_j U(\left|v_j\right\rangle)$$

and preserves the normalization constraint

$$\text{If } \sum_j \alpha_j U(\left|v_j\right\rangle) = \sum_j \alpha'_j \left|v_j\right\rangle \text{ then } \sum_j |\alpha'_j|^2 = 1$$

Unitary operators

Preservation of the normalization constraint means that unit length vectors (and thus orthogonal subspaces) are mapped by U to unit length vectors (and thus to orthogonal subspaces).

This entails a condition on valid quantum operators: they must preserve the inner product, i.e.

$$(U|v\rangle, U|w\rangle) = \langle v|U^\dagger U|w\rangle = \langle v|w\rangle$$

which is the case iff U is unitary, i.e. $U^\dagger = U^{-1}$:

$$U^\dagger U = UU^\dagger = I$$

Unitarity

- Preserving the inner product means that a unitary operator maps orthonormal bases to orthonormal bases.
- Conversely, any operator with this property is unitary.
- If given in matrix form, being unitary means that the set of columns of its matrix representation are orthonormal (because the jth column is the image of $U|j\rangle$). Equivalently, rows are orthonormal (why?)

Unitarity

Unitarity is the only constraint on quantum operators: Any unitary matrix specifies a valid quantum operator.

This means that there are many non-trivial operators on a single qubit, in contrast with the classical case where the only non-trivial operation on a bit is complement.

Finally, because the inverse of a unitary matrix is also a unitary matrix, a quantum operator can always be inverted by another quantum operator

Unitary transformations are reversible